A Report
on

IntervAI

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School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

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Batch: 2026-27

CANDIDATE'S DECLARATION

I, Parshant, do hereby declare that the internship report titled “Student Query Manager ” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Parshant

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the AICTE–EduSkills Virtual Internship Program. I am thankful for the structured learning material and guidance provided by EduSkills and AICTE under the Google AI-ML domain, and to IILM University, Greater Noida, for providing me the opportunity to undertake this internship as part of my B.Tech CSE curriculum.

I am sincerely grateful for the opportunity to apply my learning to a real project, Stdent Query Manager, and to upgrade it into a Retrieval-Augmented Generation based system during the course of this internship.

I express my gratitude to my parents for their blessings and continued support.

Signature

Student Name: Parshant

Roll No.: CS-2341768

INTERNSHIP COMPLETION CERTIFICATE



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4. Project Description

4.1 Introduction

Student Query Manager is a web-based academic query management application built using React, FastAPI, and Python. It provides separate interfaces for students, faculty members, and administrators, enabling students to submit academic queries, faculty members to manage assigned queries, and administrators to manage the system's knowledge base.

The project extends conventional query management by integrating Retrieval-Augmented Generation (RAG) for AI-powered academic question answering. Administrators can upload academic documents, which are processed, converted into vector embeddings, and stored in ChromaDB. When a student asks a question, the system retrieves relevant document content and uses the Google Gemini API to generate a context-based answer with source information where available.

The objective of the project is to combine structured academic query management with AI-powered information retrieval in a single application.

4.2 Organization Profile

This internship was completed under the AICTE–EduSkills Virtual Internship Program, a 10-week virtual internship in the "Google AI-ML" domain (January 2026 to March 2026), jointly facilitated by the Ministry of Education's National Internship Portal, the All India Council for Technical Education (AICTE), and EduSkills, with the India Edu Program supported by Google for Developers. EduSkills, which operates under the tagline "Nation Building Through Skills," partners with academic institutions such as IILM University, Greater Noida, to provide students with structured, industry-relevant, project-based virtual internships in emerging technology domains.

4.3 Problem Statement

● Traditional academic query systems often depend on manual responses from faculty members, which can delay query resolution.

● Students may repeatedly ask questions whose answers already exist in institutional documents.

● Academic information distributed across multiple documents can be difficult to locate efficiently.

● Conventional query management systems may lack semantic search and AI-powered document-based question answering.

● AI-generated answers require relevant source information to improve transparency and reduce unsupported responses.

4.4 Project Objectives

● To develop a web-based academic query management system using React and FastAPI.

● To implement JWT-based authentication and role-based access control for students, faculty, and administrators.

● To enable query submission, category-based faculty assignment, response management, and status tracking.

● To implement document upload and text extraction for PDF, TXT, and DOCX files.

● To generate semantic embeddings using Sentence-Transformers and store them in ChromaDB.

● To integrate the Google Gemini API for generating answers grounded in retrieved document content.

● To provide source information and knowledge-gap responses when relevant information is unavailable.

4.5 Scope of the Project

The scope of the project includes:

● A role-based academic query management system for students, faculty members, and administrators.

● Category-based faculty assignment and query status tracking.

● Administrative document upload, listing, and deletion functionality.

● A document-processing pipeline for text extraction, cleaning, and overlapping chunk generation.

● Semantic embedding generation using the `all-MiniLM-L6-v2` model and vector storage using ChromaDB.

● A RAG-based question-answering endpoint integrated with the Google Gemini API.

● Frontend integration for student AI queries and administrative knowledge-base management.

Some application records currently use in-memory storage, and full PostgreSQL integration and production deployment remain future improvements.

4.6 Technologies and Tools Used

● **Frontend:** React 19, Vite, Tailwind CSS 4, React Router 7, JavaScript.

● **Backend:** Python, FastAPI, Uvicorn, Pydantic, PyJWT.

● **Document Processing:** PyMuPDF, python-docx, Python text-processing utilities.

● **Embeddings:** Sentence-Transformers (`all-MiniLM-L6-v2`), generating 384-dimensional vectors.

● **Vector Database:** ChromaDB, configured for cosine-distance-based retrieval.

● **Large Language Model:** Google Gemini API for context-grounded answer generation.

● **Testing and Development:** Pytest, ESLint, Visual Studio Code, PowerShell, and Vite build tools.

4.7 System Architecture

Student Query Manager V2 follows a frontend–backend architecture in which the React frontend communicates with the FastAPI backend through REST APIs. The backend handles authentication, role-based authorization, query management, document processing, and AI question answering.

The RAG pipeline follows this flow: an administrator uploads a document; the backend extracts and cleans its text, divides it into overlapping chunks, generates embeddings using Sentence-Transformers, and stores them in ChromaDB. When a student submits a question, the system retrieves relevant chunks and passes sufficient context to the Gemini API to generate an answer. The response includes available source information, while insufficient retrieval results trigger a knowledge-gap respons

4.8 Methodology

The project was developed incrementally, beginning with backend API organization, authentication, role-based access control, and academic query management. The React frontend was integrated with backend endpoints for authentication, queries, and administrative functionality.

The RAG pipeline was then implemented through separate services for document extraction, text cleaning, chunking, embedding generation, vector storage, retrieval, and language-model interaction. PDF, TXT, and DOCX processing was added, with source metadata maintained where available.

Finally, the backend and frontend were tested independently. The recorded backend test run passed 19 tests with two warnings, the frontend lint check completed without warnings or errors across 28 files, and the Vite production build succeeded. Further testing is required to evaluate retrieval accuracy and the factual correctness of generated answers.

4.9 Expected Outcomes

Student Query Manager V2 aims to simplify academic query handling and improve access to institutional information. The system provides structured query management, role-based access, category-based faculty assignment, and document-based AI question answering.

The RAG integration enables students to obtain answers based on relevant uploaded documents, with source information available for verification. The project also establishes a foundation for future improvements, including persistent database integration, enhanced retrieval accuracy, asynchronous document processing, and production deployment.

4.10 Certificates of Completion and Other Communication/Mail Proof

The Internship Completion Certificate is attached in Section 3 of this report. The internship offer letter, issued by EduSkills, is attached below as proof of enrolment in the AICTE–EduSkills Virtual Internship Program.

EduSkills
Nation Building Through Skills

Ref No: C15Y2026M101350215

Internship Offer Letter

Student Name	: Parshant Sharma
Stream	: B.Tech
Branch	: B.Tech(Artificial Intelligence (AI) and Data Science)
Institute Name	: IILM University, Greater Noida
Internship Domain	: Google AI-ML
Internship Duration	: January 2026 - March 2026
AICTE Student ID	: STU6659ba83e4fe21717156483

We are pleased to inform you that you have been selected for the 10-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

EduSkills®
Nation Building Through Skills

The week wise Structure of the Internship is as Follows:

Week	Module	Description
week 1	Program neural networks with TensorFlow	Everything you need to know to demystify machine learning, from the first principles in the new programming paradigm to creating convolutional neural networks for advanced image recognition and classification that solve common computer-vision problems.
week 2	Get started with object detection	Basics of object detection and how to integrate a pretrained object detector into your mobile app.
week 3	Go further with object detection	Train your own custom object-detection models using TensorFlow Lite and the TensorFlow Lite Model Maker library, and build on all the skills you gained in the Get started with object detection pathway.
week 4	Get started with product image search	Build a product image search feature for a mobile app using on-device Object Detection.
week 5	Get started with product image search	Detect objects in static images and live camera feeds on Android to build a visual product search.
week 6	Go further with product image search	Build and call a product image search backend from a mobile app, enhancing the product search feature you built in the Get started with product image search pathway.
week 7	Go further with image classification	Build custom image-classification models and improve the skills you gained in the Get started with image classification pathway.
week 8	Go further with image classification	Create a custom image classifier model and integrate it into your app.
week 9	Assessment	Assessment
week 10	Final Credential Validation	Final Credential Validation

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.

Bibek Ranjan
Director, EduSkills

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NATIONAL INTERNSHIP PORTAL

5. Bibliography/References

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2.FastAPI Documentation — https://fastapi.tiangolo.com/

3.Tailwind CSS Documentation — https://tailwindcss.com/docs

4.PyMuPDF Documentation — https://pymupdf.readthedocs.io/

5.Sentence-Transformers Documentation — https://www.sbert.net/

6.ChromaDB Documentation — https://docs.trychroma.com/

7.Google Gemini API Documentation — https://ai.google.dev/gemini-api/docs

8.Pydantic Documentation — https://docs.pydantic.dev/

9.Pytest Documentation — https://docs.pytest.org/